Project Report

I started this project with the idea of making an AC voltage protector using Arduino Uno and a Finite State Machine. My plan was to monitor the grid voltage and automatically disconnect the load during overvoltage or undervoltage. I used SimulIDE for the simulation because I wanted to see the actual circuit behaviour along with the Arduino code.

At first, I created a 220V, 50Hz AC source and used a 44:1 transformer to step down the voltage. Since Arduino cannot directly measure AC voltage, I made a voltage sensing circuit using three resistors (100kΩ, 27kΩ and 100kΩ) and a 5V DC bias. Getting the correct voltage reading was one of my first challenges. I had to check the waveform using an oscilloscope and adjust the calibration until the Arduino reading was close to 220V.

Then I connected a relay through an NPN transistor and wrote the Finite State Machine code for normal operation, overvoltage, undervoltage and fault latching. I set the protection limits at 180V and 250V. One thing I wanted was that the relay should not turn ON automatically after a fault, even if the voltage becomes normal again. So I added a manual reset button and later included a reset command through serial communication.

After the basic protection worked, I tried to add current measurement using a 4.7Ω shunt resistor and three 10kΩ resistors. This part gave me trouble. The Arduino was showing around 0.12A while the expected current was around 0.22A. I checked the resistor connections, DC offset and oscilloscope waveforms several times, but I could not fully solve the difference. So I kept the current reading for monitoring and did not add overcurrent protection.

Another problem was connecting SimulIDE to my Python dashboard. I had to configure virtual serial ports using COM5 and COM6, and sometimes the connection stopped working when I restarted or changed the simulation. After fixing the port settings, I could monitor voltage and fault conditions from the dashboard and also change the voltage limits remotely. I also faced some wiring and oscilloscope issues while rearranging the circuit, so I checked the signals again before continuing.

In my tests, the Arduino measured around 220V during normal operation and the relay stayed ON. When I increased the voltage to around 270V, the relay turned OFF and the fault remained latched until I reset it at a safe voltage. I also tested the undervoltage condition by reducing the grid voltage to around 170V, below the 180V limit. The Arduino detected the fault and turned OFF the relay. Even after I returned the voltage to 220V, the relay stayed OFF until I manually reset it. The final simulation works for the main voltage protection functions, although the current measurement is still inaccurate. I have only tested the system in SimulIDE, not on real hardware.